
Cold Review of the Current $\mathcal{ALCI}_{\text{RCC5}}$ Development

Overview-paper audit, Lean artifact review,
and a renewed attack on full decidability

Review date: 9 September 2026

Submission reviewed. Attached archive `alcircc5-master.zip`, archive comment / commit identifier `795b2707b72e927c56b465ef0413b77aeb9a66a6`; overview paper `papers/overview_arxiv.tex` and the five canonical files under `formal/`.

Independent adversarial adequacy review

Claims are graded by what the archived sources actually establish, not by filenames, comments, or intended future bridges.

Abstract

This report begins with the 52-page overview paper and then audits the current Lean sources, executable probes, and the proposed routes to full $\text{ALCI}_{\text{RCC5}}$ decidability. The central verdict is deliberately two-part. First, the archive gives strong source-level evidence for a genuine, executable, reportedly kernel-checked decision procedure for *bare concept satisfiability* in the polarity-correct $\forall\text{PO}$ -free fragment, including the bridge to arbitrary nonempty-set semantics. Second, it does *not* prove decidability of the full logic. The current cone construction is complete in the no-false-negative direction for the full language, but its model-extraction/soundness argument uses the fragment restriction exactly at positive $\forall\text{PO}$.

The renewed attack did not resolve the problem left open by the primary 2006 comparator and unresolved in this snapshot. It did produce a sharper diagnosis: explicit unsatisfiable formulas survive the current unary-signature architecture because non-singleton RCC5 compositions impose obligations between independently generated witnesses. A converse satisfiable example shows that declaring all such cross-pairs to be PO is not a repair. The report also gives (i) a witness-generated countable-model lemma, and (ii) a two-layer encoding of ordered-disjoint frames into one strict partial order. The latter absorbs RCC5 heredity into transitivity, but introduces a synchronized fixed-point-free involution not covered by the standard two-variable/one-transitive-relation theorem. These are useful reductions, not a decision proof.

For publication, the recommendation is major rewrite: extract a focused paper on the reported fragment theorem and normal form, and recast the full-logic discussion as a research program with every missing implication stated explicitly.

Contents

1	Scope, materials, and review standard	2
2	Executive verdict	2
3	Cold review of the overview paper	2
4	Lean artifact audit	5
5	Renewed attack on full decidability	8
6	Literature and novelty audit	12
7	Recommendations and research plan	13
8	Final assessment	15
A	Artifact hashes and reproduction notes	15
B	Claim-to-evidence matrix	16

1 Scope, materials, and review standard

1.1 Snapshot and objects reviewed

The attached ZIP, rather than an assumed later repository state, is the review object. Its SHA-256 digest is

4a0415b3e3202595d1b1290e6cc741068d0423c0c2877e815b9b5fcbe96bd2a8.

The ZIP comment identifies commit `795b2707b72e927c56b465ef0413b77aeb9a66a6`. The overview source digest is `be28d3da...429e`; its included PDF has 52 pages and digest `e684c506...322a`. Full hashes of the formal sources appear in Appendix A.

I attempted to compare the public GitHub head and to obtain a fresh Lean toolchain. Network access to those endpoints was denied in this execution environment, and no `lean`, `elan`, or `lake` executable was preinstalled. Accordingly, this is a cold review of the identified archive, not a statement about commits after it, and the Lean judgment below separates source audit from independent compilation.

1.2 Four evidential levels

I use four labels throughout:

1. **Source-audited Lean declaration:** a declaration present in a canonical Lean source, with no project-specific axiom or proof escape visible in the source. Without a local build this is not itself a kernel-verification claim.
2. **Archived build claim:** a build/axiom result recorded in the submission’s transcripts, but not reproduced in this environment.
3. **Executable evidence:** a Python checker or finite exhaustive test reproduced during this review. This is regression evidence, not a proof of an unbounded theorem.
4. **Mathematical/prose claim:** a paper argument, conjecture, or proposed bridge that is not an instantiated Lean theorem.

This discipline matters here: the repository contains excellent source-audited, reportedly kernel-checked combinators whose decisive completeness hypotheses are still structure fields or theorem parameters.

2 Executive verdict

Overall verdict

Full $\text{ALCI}_{\text{RCC5}}$ decidability is not proved and was not cracked in this review. The problem is unresolved in the submitted snapshot and inspected corpus. The $\forall\text{PO}$ -free concept-satisfiability theorem is the strongest completed result and appears technically real. The overview paper, however, currently promotes an architecture-specific conjectural route (F6 plus a uniformization bridge) into a “one keystone” reduction. As a formal-results paper it requires a major rewrite; as two papers—a focused fragment/formalization paper and a candid research-program/history paper—the material is much stronger.

Confidence calibration. I found no evidence that Lean accepted a false theorem. The problem is almost the opposite: several prose claims outrun the hypotheses of sound, honest Lean theorems. The fragment result deserves a high-confidence label subject to a fresh pinned build. Any headline resolution of the full logic deserves a low-confidence label until a model-existence/completeness theorem is supplied.

3 Cold review of the overview paper

Claim	Verdict	Basis / qualification
Full concept satisfiability is decidable	Unresolved here	No complete finite-presentation theorem; current cone extraction fails with positive $\forall\text{PO}$.
Full unsatisfiability has a sound cone test	Supported	<code>coneScheme_complete</code> gives full-language no-false-negatives; the contrapositive theorem is explicit.
$\forall\text{PO}$ -free concept satisfiability is decidable	Strongly supported	End-to-end source chain from raw polarity test through arbitrary nonempty-set semantics; clean external rebuild still needed.
Normal form / set representation	Supported	Substantive Lean development, but the overview cites the wrong file for the converse assembly.
F6 is the sole remaining theorem	Not established	F6 is informal and the $\text{F6} \Rightarrow \text{uniformization} \Rightarrow \text{finite-checker-completeness}$ bridge is open.
Overview ready as a formal-results paper	No	Major claim calibration, status consolidation, scope, citation, and reproducibility changes are required.

Table 1: Current-state verdict at archive commit 795b2707...66a6.

3.1 What is strong

The paper has real strengths worth preserving:

- It records the long sequence of failed proof routes more candidly than most research narratives and often distinguishes abstract relation- algebra semantics from topological semantics correctly.
- The ordered-disjoint normal form is a useful conceptual compression: PP becomes strict order, DR becomes hereditary conflict, and PO is the residual incomparable non-conflict relation.
- The final cone-scheme section points to a substantial Lean theorem reported to build for the whole polarity-correct $\forall\text{PO}$ -free fragment, not merely the older three “pure quadrants.”
- The paper exposes counterexamples and retractions instead of deleting them from the historical record.
- The abstract-to-set satisfiability bridge and the explicit theorem-name map are exactly the right reproducibility instincts, even though the map needs correction.

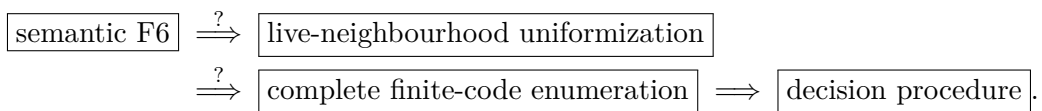
3.2 Ranked findings

ID	Severity	Finding	Required correction
O1	Major	“Reducing it to one keystone” is not a proved reduction. F6 is informal, and the finite-code completeness theorem is not derived from it.	Retitle and display every missing arrow; call F6 a candidate sufficient route within one architecture.
O2	Major	The identity-selector obstruction is described as an iff / theorem-level characterization although forceability by concepts is not proved.	Reclassify it as a candidate obstruction and finite-network diagnostic.
O3	Major	The current-status table contains both the superseded “three quadrants plus mixed witness” state and the later whole-fragment theorem.	Keep one current row; move the older result to a chronology.
O4	Major	Scope is easy to overread: the result is bare concept satisfiability, not general TBoxes or unrestricted subsumption. The raw fragment is not negation-closed.	Put the restriction in the abstract and theorem box; state when $C \sqcap \neg D$ leaves the fragment.

ID	Severity	Finding	Required correction
O5	Major	“No false positive or false negative ever occurred” conflicts with the disclosed retracted quasimodel’s false rejection and conflates labelled regressions with oracle comparisons.	Name the current implementation and corpus; separate 99 labelled cases from the remaining comparison runs.
O6	Major	The artifact map attributes both normal-form directions to <code>RCC5NormalForm.lean</code> ; the final frame and set assembly is in <code>POFreeLift.lean</code> .	Cite <code>odNet_frame</code> , arbitrary-set interpretation, and <code>satisfiable_iff_set</code> at their actual locations.
O7	Major	The Π_1^0 /dovetail row says “Lean-certified,” but the RCC5-specific completeness and formal proof enumeration are assumptions, not an instance.	Label only the generic race combinator source-audited and reportedly kernel-checked.
O8	Moderate	Failure of known grid constructions is sometimes stated as “no concept can enforce a grid.”	Say “no known encoding” unless a genuine non-definability theorem is proved.
O9	Moderate	Patchwork is glossed as triple/path consistency implying global consistency and rhetorically credited with resisting grids.	Separate overlap amalgamation from path-consistency completeness and cite the exact theorem used.
O10	Moderate	“Every load-bearing claim mechanically checked” and “local algebra exhausted” exceed the evidence.	Enumerate source-audited statements; label F6, W2’, selector forceability, and automatic-cover completeness conjectural.
O11	Moderate	Novelty and prior-art framing need tightening: abstract RCC5 set representability predates this development; the plain concrete-domain contrast is too categorical.	Credit relation-algebraic representation results and compare the role-forming predicate operator separately from plain $\mathcal{ACC}(D)$.
O12	Minor	Review counts drift (17 versus 20), mutable <code>master</code> URLs are used, the overview repeats superseded material, and the GIS example turns membership into equality without a nominal.	Pin a release/commit, add CI/build manifest, reconcile counts, shorten the polemical appendix, and correct the example.

3.3 The missing dependency chain

The overview’s central editorial problem is visible in one dependency chain:



The last implication is formalized in conditional form. The first two are not. The theorem `decidableSat_of_finScheme` in `Round19Transport.lean` consumes an explicit enumeration and a `complete` hypothesis. In `ForcingReduction.lean`, the RCC5-specific fact is supplied through an `adequacy` field, while its `F6fin` is qualitative existence, not an algorithm computing a bound from the input concept. The width-barrier prose itself says that the coarse W2’ statement is false and proposes a finer uniformization lemma. Calling only F6 “the remaining theorem” therefore hides a theorem-level bridge.

There is also a weaker, unbounded dovetail route. If finite certificates have a computably enumerable syntax, their checker is decidable and sound, and every satisfiable input has *some* accepted code, then no

formula-computable size bound is needed: enumerate certificates in parallel with first-order refutations. The snapshot formalizes this abstract race but neither instantiates the RCC5 certificate-completeness premise nor formalizes the refutation side. Thus “no computable bound” is fatal only to the direct finite-search route; the deeper gap in both routes is the unproved RCC5 regularization/adequacy theorem for one exact effective certificate syntax.

There is a harmless generic sense in which decidability is equivalent to an effective complete certificate relation. It does not follow that this particular live-width F6 is necessary, or that failure of this architecture would imply undecidability. The paper should use “remaining gap in this architecture,” not “the irreducible core of the problem.”

3.4 Scope of the reported fragment theorem

After normalization, the fragment forbids positive $\forall\text{PO}$. On raw syntax the polarity-correct predicate also forbids a negative $\exists\text{PO}$, because normalization turns it into $\forall\text{PO}$. It permits arbitrary existential modalities, including $\exists\text{PO}$, and all other universal RCC5 modalities.

This is an expressive concept-satisfiability fragment, but it is not closed under negation and is not automatically stable under the usual reduction of subsumption to unsatisfiability of $C \sqcap \neg D$. General TBox internalization can likewise introduce $\forall\text{PO}$. These limits do not diminish the theorem; stating them prominently makes it a credible submission candidate after the build and literature prerequisites in Section 7 are met.

3.5 Recommended publication split

Paper A: focused formal result

Lead with the ordered-disjoint normal form, the cone control space, full- language completeness, fragment soundness, the raw polarity theorem, and the abstract/arbitrary-set-semantics equivalence. Include a pinned build, axiom footprint, complexity bound (or an honest statement that it remains loose), and a compact comparison with prior RCC5 representation work. This appears to be a substantial paper after an external clean Lean build and literature sweep.

Paper B: research program and methodology

Present F6, selector forceability, mosaics, automatic presentations, and the history of failed routes as conjectures and experimental evidence. This can be valuable without claiming that the full decision problem has been reduced to one proved keystone. Move superseded theorem tables and the long AI-methodology discussion here.

4 Lean artifact audit

4.1 Canonical inventory

The five canonical files total 51,322 lines. The archive also contains many historical copies, excerpts, and review bundles; those should not be allowed to blur which files constitute the release theorem.

4.2 Proof-escape and axiom scan

Across `formal/*.lean`, the static audit found:

- no actual `sorry` or `admit`;
- no project-specific `axiom` declaration (the sole apparent hit is prose inside a block comment);
- no `unsafe`, `opaque`, `partial`, `extern`, or `implemented_by`;
- 910 `#print axioms` commands in the monolithic fragment file;

File	Lines	SHA-256 prefix	Actual role
<code>P0FreeLift.lean</code>	45,375	5f85ff556d0d	Current cone scheme, fragment decision chain, raw syntax, and final arbitrary-set-semantics bridge.
<code>Round19Transport.lean</code>	5,309	60d08f4c3adc	Older certificate-to-model soundness and conditional finite-scheme decider.
<code>SemiDecidability.lean</code>	262	8a4346385619	Generic dovetail schema; no instantiated RCC5 certificate/refutation system.
<code>RCC5NormalForm.lean</code>	231	438e82778910	Compact forward normal-form and core η -representation lemmas; not the whole converse assembly.
<code>ForcingReduction.lean</code>	145	37f18d9b76f3	Abstract reduction under an RCC5-specific adequacy field; not a proof of adequacy or effective F6.

Table 3: Canonical Lean sources in the submitted archive.

- no `native_decide` use in `P0FreeLift.lean`; 28 occurrences in `Round19Transport.lean` are diagnostics/non-vacuity examples, and two in `SemiDecidability.lean` are toy examples.

The archived transcripts report successful Lean 4.32.0/4.33.1 builds and the standard footprint `propext`, `Classical.choice`, and `Quot.sound` for the fragment capstones. Since this review could not run Lean, that sentence is an archived build claim, not an independently reproduced kernel check. The source scan nevertheless found no hidden problem-specific assumption in the canonical cone theorem chain.

4.3 What the current fragment chain really certifies

Declaration	Line	Role
<code>coneScheme_complete</code>	42317	From any full-language model/root, obtain a surviving cone signature. No <code>P0Free</code> premise.
<code>unf_truth</code>	44006	Truth lemma for the fresh-occurrence model; <code>P0Free</code> is used precisely in the $\forall$ PO branch near 44048.
<code>coneScheme_sound</code>	44093	A survivor yields a model under the fragment restriction.
<code>coneScheme_correct_at</code>	44179	Finite acceptance iff satisfiable for the fragment.
<code>decidableSat_cone</code>	44196	Executable NNF decision procedure.
<code>coneScheme_unsat_full</code>	44217	Sound full-language rejection only: finite elimination implies unsatisfiable.
<code>nnfP_correct</code>	44452	Raw syntax and negation-normal-form bridge.
<code>FP0Free</code>	44530	Polarity-sensitive fragment predicate on raw formulas.
<code>decidableFSat</code>	44589	Raw-formula fragment decider.
<code>satisfiable_iff_set</code>	44948	Abstract RCC5 satisfiability iff arbitrary nonempty-set satisfiability.
<code>decidableSetSat</code>	44976	Set-semantics fragment decision instance.

The clean summary is therefore:

$$\underbrace{\text{full-language completeness}}_{\text{no false rejections}} + \underbrace{\text{fragment model extraction}}_{\text{no false acceptances}} = \text{fragment decidability.}$$

For the full language, only the left half is present.

4.4 What the other full-logic files do not supply

Conditional finite scheme. `Round19Transport.lean` proves a credible certificate-to-model soundness pipeline and a fixed finite checker’s soundness. Its final `decidableSat_of_finScheme` theorem takes a `complete` argument. No term inhabiting the RCC5 completeness premise is supplied.

Dovetailing. `SemiDecidability.lean` proves that a generic `DovetailScheme` whose certificate and refutation sides are both sound and complete can be raced. Those four properties are fields. There is no formal first-order syntax, Goedel coding, proof enumeration, completeness theorem, or $\text{ALCI}_{\text{RCC5}}$ instance. Thus the mathematical $\text{co-r.e.}/\Pi_1^0$ placement may be right, while the end-to-end certification label is not.

Forcing. `ForcingReduction.lean` is correct abstract order theory under the `adequacy` field. It does not build an adequacy instance for RCC5 presentations. Its formal `F6fin` is qualitative, despite an “effective” comment: this is insufficient for the direct bounded-search theorem, though it could support dovetailing after an actual enumerable certificate language, checker-completeness proof, and refutation enumerator are supplied.

4.5 Reproduced executable evidence

Probe	Reproduced scope	Result
wp84	1,048,576 raw two-point and 1,953,125 functional three-point structures; 1,050 standard-translation triples	Zero mismatches
wp88	36,000 ordered-disjoint networks through size nine	All five relations reproduced by η ; zero failures
wp133	290 fresh-occurrence prefixes	Zero frame violations; reported truth residues attributed to omitted controls
wp134	Seven curated elimination diagnostics	All expected outcomes
wp135	2,415 fragment and 2,431 full satisfiable instances plus mutations/boundary controls	All tested completeness obligations passed
wp87	243 random/curated concepts, including depth five	Two reasoners agreed; zero reported disagreements
Cover-tree stress	911 tests (50 SAT, 13 UNSAT, 36 adversarial, 512 triples, 300 deeper random tests)	911/911; zero mismatch

Table 5: Finite regression evidence reproduced in this review.

These runs materially increase confidence in definitions and regressions. They cannot establish an unbounded model theorem, and some comparisons share the same RCC5 table or expected labels. They should be advertised as reproducible testing, not as independent proof.

5 Renewed attack on full decidability

5.1 The ordered-disjoint core

The normal form may be stated as a structure $(X, <, \#)$ where $<$ is a strict partial order, $\#$ is symmetric, irreflexive, disjoint from comparability, and downward hereditary:

$$x < y \wedge y \# z \implies x \# z,$$

with the symmetric-coordinate version following from symmetry. Then

$$\begin{aligned} \text{PP}(x, y) &\iff x < y, & \text{PPI}(x, y) &\iff y < x, \\ \text{DR}(x, y) &\iff x \# y, & \text{EQ}(x, y) &\iff x = y, \\ \text{PO}(x, y) &\iff x \neq y \wedge x \not\prec y \wedge y \not\prec x \wedge \neg(x \# y). \end{aligned}$$

Throughout, $X \neq \emptyset$ and strong EQ means $\text{EQ}(x, y) \leftrightarrow x = y$. In the submitted set bridge, distinct points are represented injectively by nonempty sets, with PP as proper subset, PPI as proper superset, DR as empty intersection, and PO as nonempty intersection with neither set containing the other. This is arbitrary nonempty-set semantics; it is not by itself a claim about one fixed full class of regular-closed regions or boundary-sensitive RCC semantics.

Relation to the original split-forest idea. The ordered-disjoint normal form is best understood as the exact semantic core of the split-forest picture, rather than as a competing construction. Its strict order is the vertical PP/PPI nesting axis; its hereditary conflict relation is the horizontal DR relation and precisely formalizes the downward “shadow” propagation used by the split-forest argument; and PO is the remaining horizontal relation once equality, comparability, and conflict have been excluded. The difference is proof-theoretic strength. An ordered-disjoint structure may still have an arbitrary partial order, including diamonds and several incomparable superparts. A split forest additionally tries to replace that order by a formula-relative tree cover, splitting join points into occurrence copies and controlling their horizontal interfaces by finite profiles, patchwork, or a quotient. Thus the certified normal form establishes the algebraic substrate, but it does not establish the bounded-width split-cover theorem needed for the original full-logic decision route. The obstruction is especially sharp for $\forall\text{PO}$: splitting or unravelling can create new pairs that are neither comparable nor disjoint and hence become residual PO-neighbours, which universal PO formulas can observe. In the $\forall\text{PO}$ -free fragment those extra residual edges cannot violate a universal PO obligation, explaining why the certified fresh-occurrence unfolding retains the split-forest shape without requiring its older quotient and bounded-interface machinery. For the full logic, the missing bridge is a formula-relative split-cover theorem for ordered-disjoint structures that preserves all $\forall\text{PO}$ obligations.

The full problem is therefore modal satisfiability over total five-way labellings induced by an order with hereditary conflict. The hard part is not choosing one witness; it is coordinating the residual PO relation among many witnesses after order and conflict closure.

5.2 Four boundary formulas

The following formulas are small enough to become permanent formal regressions. They distinguish three inadequate repairs.

5.2.1 A direct $\forall\text{PO}/\exists\text{PO}$ clash

$$C_{\text{dir}} = \forall\text{PO}.A \sqcap \exists\text{PO}.\neg A.$$

This is immediately unsatisfiable: the existential witness is a PO-neighbour and must satisfy both A and $\neg A$. When the current cone control is run with its $\forall\text{PO}$ information erased, the finite control survives and accepts. Thus simply reusing fragment extraction for the full language is impossible.

5.2.2 A cross-branch composition clash

$$C_{\text{joint}} = \exists \text{PP}.(\forall \text{PPI}.A \sqcap \forall \text{PO}.A) \sqcap \exists \text{PO}.\neg A.$$

Let y be the PP-witness and z the PO-witness from the root x . Then $y\text{PPI}x$ and $x\text{PO}z$, so the RCC5 table forces $\text{rel}(y, z) \in \{\text{PPI}, \text{PO}\}$. Either universal at y forces $A(z)$, contradicting $\neg A(z)$. Independent witness births miss this joint obligation.

5.2.3 A non-singleton-composition clash with no $\exists \text{PO}$

Let

$$\begin{aligned} C_0 &= \exists \text{PPI}.(\exists \text{PP}.A) \sqcap \forall \text{PP}.\neg A \sqcap \forall \text{PPI}.\neg A \sqcap \neg A, \\ C_{\text{bad}} &= C_0 \sqcap \forall \text{PO}.\neg A. \end{aligned}$$

If $y\text{PP}x$ and $y\text{PP}z$ with $A(z)$, then

$$\text{rel}(x, z) \in \text{comp}(\text{PPI}, \text{PP}) = \{\text{EQ}, \text{PP}, \text{PPI}, \text{PO}\}.$$

Under the submitted strong-EQ, nonempty-region semantics, the composition entry excludes DR. The four root constraints eliminate its four labels, so C_{bad} is unsatisfiable. The archived diagnostic reports a three-signature post-fixed family for the current unary cone architecture: the x, z pair is constrained by a non-singleton composition entry and is never emitted as an explicit $\exists \text{PO}$ demand. I did not independently reconstruct that exact family in this run, so its survival should be treated as a source/prose diagnostic; the semantic unsatisfiability argument itself is direct. In any case, strengthening only explicit PO-witness transitions does not address this kind of hidden composition obligation.

5.2.4 A satisfiable guard against “make all cross-pairs PO”

$$C_{\text{sat}} = \exists \text{PP}.(A \sqcap \forall \text{PO}.A) \sqcap \exists \text{PP}.\neg A.$$

Take a strict chain $x < z < y$, use y for the first witness and z for the second, and put A at y and $\neg A$ at z . The witnesses must be distinct. At y , the $\forall \text{PO}.A$ condition is vacuous for these points because $z < y$. Thus C_{sat} is satisfiable. Any blanket safety rule requiring distinct independently created witnesses to be pairwise PO-compatible would reject it.

Consequence for the current cone architecture

A successful full construction must somehow account for relation choices and triangle obligations among contemporaneous witnesses; an explicit joint matrix is the most direct route. The current endpoint types and per-demand target tests do not do so, while a universal “fresh pairs are PO” default is ruled out by C_{sat} . These examples refute the submitted extraction and the named naive repairs; they do not prove that every possible unary-state or local-compatibility refinement must fail.

5.3 Why ordinary finite types are not enough

No finite cardinal bound. The concept

$$C_{\infty} = \exists \text{PP}.\top \sqcap \forall \text{PP}.\exists \text{PP}.\top$$

forces an infinite ascending strict-PP chain. Therefore a finite-model certificate cannot be the target of a model-certificate route. Such a route must be able to represent infinite models finitely; this observation does not exclude a fundamentally different decision method.

Filtration collapses strictness. Repeated points on the chain can have the same closure type. Identifying them turns PP into equality or creates a PP self-loop. Unrolling a quotient does not automatically help: new pairs between copies receive residual PO, which can violate positive $\forall\text{PO}$ obligations.

Spectra do not determine horizontal incidence. Even profiles enriched with lower/upper type spectra fail to record the DR/PO matrix. For every n , take

$$X_n = \{b\} \cup \{L_i, U_i, A_i : 0 \leq i < n\},$$

order only $b < U_i$, and let conflict be the symmetric closure of $b \# L_i$ and $L_i \# A_i$. Give all members of each named family the same unary type. Every family then has a uniform lower/upper spectrum, while

$$L_i \text{DRA}_j \iff i = j, \quad L_i \text{POA}_j \iff i \neq j.$$

A quotient whose conflict choice factors only through profiles must smear the diagonal or lose a required edge. The example does *not* prove undecidability: a fixed concept might have a more regular alternative model. It motivates one candidate regularization lemma—a formula-bounded continuation index for conflict rows, or an equally effective finite presentation—without proving that this is the unique possible route.

5.4 A witness-generation base lemma

Witness-generated countable-model property

If a pointed full $\mathcal{ALCI}_{\text{RCC5}}$ concept is satisfiable, it has a countable induced submodel generated by choosing one fixed witness for every true existential closure formula at every retained point. If the closure contains k existential formulas, the chosen witness-generation graph has branching at most k . The semantic RCC5 graph remains total, so this is not a bounded-degree model theorem.

Proof. Fix once and for all a selector $w(x, E)$ choosing one original-model witness for every pair consisting of a point x and a true existential formula E from the NNF closure at x . With W_0 the satisfying root, put

$$W_{m+1} = W_m \cup \{w(x, E) : x \in W_m, E \text{ is true and existential at } x\}.$$

The union over ω is countable, and each point has at most k outgoing edges in this chosen generation graph. Restrict all predicates and relations to that union. The RCC5 frame conditions are universal, hence inherited by an induced substructure. Atomic literals are unchanged; both conjuncts of a true conjunction and one disjunct of a true disjunction remain true; universal formulas only become easier after restriction; and each true existential has a selected witness. Induction over NNF formulas preserves truth at retained points, in particular at the root. $\square$

This lemma is Lean-friendly and should be formalized. It is only base camp. A recursive finitely branching tree may have no computable path, much less an automatic one. Countability, compactness, and finite branching do not supply the required regular model.

5.5 A two-layer partial-order encoding

The review found a concise way to absorb conflict heredity into ordinary transitivity. Given $(X, <, \#)$, form two copies Lx, Rx and define a binary relation Q by

$$\begin{aligned} Lx Q Ly &\iff x < y, & Rx Q Ry &\iff y < x, \\ Lx Q Ry &\iff x \# y, & Rx Q Ly &\iff \perp. \end{aligned}$$

Lemma. Q is a strict partial order.

Proof sketch. Irreflexivity of Q follows from strictness of $<$ on each same-sort copy; the copies are disjoint, so a cross edge is never a Q -self-loop. The all-left and all-right transitivity cases are order transitivity. The mixed case $Lx Q Ly Q Rz$ is exactly $x < y \wedge y \# z \Rightarrow x \# z$; the case $Lx Q Ry Q Rz$ is the symmetric-coordinate hereditary law. No relation leaves the right copy for the left, so there are no other cases. $\square$

Let f swap Lx and Rx . On the left copy one recovers

$$x < y \iff Q(x, y), \quad x \# y \iff Q(x, f(y)),$$

and hence defines PO as the residual. The original two-variable standard translation now uses one transitive relation Q , unary sort predicates, and the matching involution f .

Why this is not yet a decidability reduction

For an exact two-copy image, the target class must enforce an involution $f^2 = \text{id}$ that swaps the left and right sorts, prohibit right-to-left Q -edges, and require, for all left-sort x, y ,

$$Q(x, y) \leftrightarrow Q(f(y), f(x)), \quad Q(x, f(y)) \leftrightarrow Q(y, f(x)), \quad \neg Q(x, f(x)).$$

The last condition is the image of conflict irreflexivity. Standard decidability of two-variable first-order logic with one transitive *relational* symbol does not automatically allow a unary function or an FO-definable perfect matching. Replacing f by its graph makes functionality a three-variable condition in plain FO^2 . Thus the construction gives a precise literature target and perhaps a new proof route, but not a theorem that can currently be invoked.

The encoding is still useful. It packages the three-variable-looking RCC5 absorption law into transitivity and isolates synchronization, rather than spatial composition, as the remaining issue. A guarded/two-sorted decision procedure specialized to this coherent involutive subclass would settle the problem.

5.6 Most credible positive route: automatic or mosaic covers

An effective certificate should carry more than unary types. Two compatible formulations are worth pursuing.

Automatic presentation. Use a regular language of point addresses, finite-state unary type map, synchronous relations for PP and DR, and regular witness relations for each existential demand. Define PPI by converse, EQ by identity, and PO residually. Given such a finite presentation, automata operations can check strictness, transitivity, conflict symmetry/irreflexivity, heredity, endpoint modal constraints, and total witness service. Soundness can then feed the reportedly kernel-checked set representation.

Pair-matrix mosaics. A finite state should contain a complete RCC5 matrix among one representative for each simultaneously live demand, with interfaces recording continuation requirements. A greatest fixed point must prove both directions:

1. mosaics realized in a model survive elimination; and
2. every surviving compatible mosaic family unfolds to one coherent ordered-disjoint model satisfying all $\forall\text{PO}$ obligations.

The second statement is where bounded joint width or finite continuation index reappears. Calling it “F6” does not prove it.

The sharp qualitative open statement is therefore

$$\text{EnumerableCover}(C) : \quad C \text{ satisfiable} \implies \text{some finite automatic (or mosaic) code passes the fixed effective checker.}$$

Together with a formal enumerable refutation side, this is enough for decidability by dovetailing. A stronger computable bound on code size would instead give a direct finite search. No proof of either cover statement is present. A useful intermediate fragment would bound the number of distinct DR rows between equal closure profiles and show that the bound is preserved by witness generation.

6 Literature and novelty audit

6.1 What the primary comparator establishes

Lutz and Wolter’s 2006 paper is the closest primary comparator [1]. The archived PDF confirms the exact citation as LMCS 2(2:5), not 2(2:2). Their representation theorems identify general RCC5 structures

with the relevant substructure semantics; their undecidability theorem applies to supremum-closed/full region classes, not to the general substructure class matching the submitted target. They prove recursive enumerability of the corresponding validity logic and explicitly leave the general RCC5 case open. Consequently, their result does not supply either full decidability or undecidability here.

The co-r.e./ Π_1^0 satisfiability upper placement used in the overview is consistent with this: validity is r.e., so unsatisfiability of a concept can be enumerated after the standard translation. What is missing from Lean is the actual formal proof system/encoding and its completeness, not the abstract idea of dovetailing two semidecidable.

6.2 Representation and concrete-domain context

Abstract/general RCC5 set representability is not new as a bare mathematical fact. The relation-algebraic literature of Duentzsch, Wang, and McCloskey [2], Li and Ying’s analysis of models and the composition table [3], and Lutz–Wolter’s representation results must be credited. The potentially novel contribution here is the particular ordered-disjoint formulation, its constructive η representation, and the reportedly kernel-checked end-to-end use in a description-logic fragment.

The overview should also narrow its claim that concrete-domain formalisms cannot form relations. Plain $\mathcal{ALC}(D)$ is indeed feature-mediated and two-sorted, but Haarslev, Lutz, and Moeller explicitly introduced a role-forming predicate operator over feature chains [4]. That formalism is still not the primitive, single-sorted, total RCC5 role partition studied here; it is simply a closer comparator than the overview currently admits.

6.3 Event structures and the doubled-poset target

The ordered-disjoint frame has the same order/conflict signature and a dual heredity law as an event-structure skeleton: event causality corresponds to the *reverse* of $< = \text{PP}$, hereditary DR is conflict, and residual PO is concurrency [5, 6]. An arbitrary frame here need not be a prime event structure: classical definitions commonly require finite causal past, which is absent from the submitted frame axioms. No direct decision theorem for this event-to-event modal signature was located in the inspected corpus; standard event-structure logics often reason over configurations and enabling transitions instead. The connection nevertheless supplies mature language for unfoldings, regular event structures, and finite-state presentations, and deserves a precise comparison.

Szwast and Tendera prove decidability of function-free relational FO^2 with one designated transitive relation and auxiliary relational predicates [7]. The theorem is a comparator, not a corollary: the direct $(<, \#)$ theory needs the three-variable hereditary axiom, while the two-layer encoding needs a synchronized unary involution/perfect matching. This exact enriched signature is the right literature question.

6.4 Search limitation

The supplied archive contains the primary 2006 paper and a substantial related-work record. No inspected primary source in that corpus settles full $\mathcal{ALCI}_{\text{RCC5}}$ satisfiability. Live web/database searching was attempted but was not available from this run, so this report does *not* certify the stronger bibliographic statement “still open as of September 2026.” Before a novelty or priority claim is published, run and record searches in DBLP, MathSciNet, zbMATH, Scopus/Web of Science, Google Scholar, and citation graphs for the 2006 paper, including terms for RCC5 modal logic, event structures, one transitive relation with functions, and regular/automatic models.

7 Recommendations and research plan

7.1 Priority actions

Priority	Action	Acceptance criterion
P0	Freeze a reproducible Lean release	Add <code>lean-toolchain</code> , a minimal Lake project, shared base module, and CI that builds the five canonical targets, rejects <code>sorry</code> /custom axioms, and records expected axiom footprints.
P0	Correct the overview's theorem status	One current fragment theorem; F6/uniformization/checker arrows shown separately; no iff selector claim; bare concept/TBox/subsumption scope explicit.
P0	Add full-boundary regressions	Formalize C_{dir} , C_{joint} , C_{bad} , and C_{sat} , including survival/failure statements for candidate controls.
P1	Extract a trusted shared core	One imported definition of atoms, composition, syntax, semantics, and ordered-disjoint representation; eliminate large copied tables across canonical files.
P1	Formalize countable witness generation	A carrier-polymorphic theorem constructing the induced countable submodel and proving NNF truth preservation.
P1	Build certificate soundness first	Define an automatic or pair-mosaic certificate in finite syntax; prove its checker decidable and its accepted objects yield models, without hiding completeness in a structure field.
P1	State the hard hypothesis honestly	A named <code>EnumerableCover</code> theorem for one effective finite-code grammar; separately label any stronger formula-computable bound used for direct search.
P2	Attack bounded-row fragments	Prove completeness when each closure-profile pair has at most m distinct hereditary-conflict rows; determine whether m can be bounded from the formula.
P2	Develop the doubled-poset route	Formalize the two-copy equivalence and survey/derive decidability for FO^2 over one strict order with the required coherent involution.
P2	Run a reproducible literature review	Publish queries, dates, databases, inclusion criteria, and a citation graph; have a human domain expert verify the priority conclusion.

7.2 Go/no-go milestones for full decidability

1. **Semantic regression gate.** Any new local rule must reject C_{bad} and accept C_{sat} . Failure means the architecture is not ready for a completeness proof.
2. **Finite-code gate.** Certificates must be inductive finite data with a terminating checker. Higher-order functions or an unbounded semantic labelling are not decision certificates.
3. **Soundness gate.** Prove checker acceptance produces an ordered-disjoint model satisfying the formula, with no completeness field in the soundness statement.
4. **Regularization gate.** Prove that every satisfiable formula has an accepted finite code in that exact effective syntax (or a stronger formula-computable bound if direct finite search is claimed). This is the decisive theorem for this positive route. If it fails, report the countermodel pattern rather than

shifting the premise to another abstraction.

5. **End-to-end gate.** Instantiate the executable decision theorem, print its axioms, run it on all boundary formulas, and reproduce it from a clean tagged checkout in CI.

7.3 What not to infer

Three negative conclusions would be premature:

- Failure of the unary cone scheme does not imply undecidability.
- Selector matrices in arbitrary finite RCC5 frames do not show that one fixed concept forces unbounded selectors.
- A countable finitely branching witness construction does not imply an automatic or computable model.

Conversely, the absence of a successful grid reduction does not imply decidability. The honest frontier is a regularization theorem or a genuinely faithful undecidability encoding.

8 Final assessment

Final answer

The attached work has crossed an important line: the $\forall\text{PO}$ -free fragment is no longer merely an experimental conjecture but is backed by a detailed, apparently escape-free Lean theorem chain and substantial regression evidence. That is the result to develop into a focused submission after a pinned independent build and current novelty review.

The full $\text{ALCI}_{\text{RCC5}}$ decision problem is unresolved in this snapshot. The submitted cone extraction is not repaired merely by erasing or adding the endpoint-only $\forall\text{PO}$ checks considered here; the formulas in Section 5.2 show why. The most credible positive target is an enumerable joint-pair/mosaic or automatic-presentation cover theorem (with a computable bound if direct finite search is desired). The two-layer partial- order encoding offers a second, sharply stated route, but the required matching involution is a nontrivial enrichment not covered by the standard FO^2 theorem.

Publication verdict on the overview: **major rewrite / reject in current formal-results form.**

Publication verdict on a separated fragment/formalization paper after a pinned independent Lean build: **promising and likely publishable, subject also to a current novelty review.**

Research verdict on the full problem: **unresolved in the reviewed snapshot and corpus; a sufficient global regularization theorem for the leading positive route is now more precisely stated, but not proved.**

A Artifact hashes and reproduction notes

Artifact	SHA-256
<code>alcircc5-master.zip</code>	4a0415b3e3202595d1b1290e6cc741068d0423c0c2877e815b9b5fcb e96bd2a8
<code>papers/overview_arxiv.tex</code>	be28d3dace3a1cf0bacbaaa253872c38e76c32a9995d3e21c70140b8 e656942e
<code>papers/overview_arxiv.pdf</code>	e684c506ac501219a93cf1b733d52dfcaa3a5c63f983bf6a65119900 0440322a
<code>formal/P0FreeLift.lean</code>	5f85ff556d0d67205432311f136a90335b4a0540576604ec0f395e4a a3db28ae
<code>formal/Round19Transport.lean</code>	60d08f4c3adcd44b90820d764911750cc3e19072a7e84e6e87174e4a 36c82678

Artifact	SHA-256
<code>formal/SemiDecidability.lean</code>	8a43463856193ca6ad053498502c17239fff8c9d33d7aab594e4f5e9d1077347
<code>formal/RCC5NormalForm.lean</code>	438e8277891054c82050d2d1c83065100881b0630692b72c7c71a5ee80d127ad
<code>formal/ForcingReduction.lean</code>	37f18d9b76f38461dd310a44adc768216af0070a88bda0dd482ce761269b3351

The following classes of checks were run from the extracted archive:

- exact hash, line-count, and forbidden-token scans over canonical Lean;
- theorem-location/dependency inspection around the cone capstones;
- `wp84_fo_pi01_transcription.py` and `wp88_canonical_representation.py`;
- `wp133`, `wp134`, `wp135`, and `wp87`;
- `src/stress_test_cover_tree.py`;
- direct execution of the full-logic boundary script showing acceptance of C_{dir} and C_{joint} when $\forall \text{PO}$ obligations are erased.

No fresh Lean compilation was possible for the reasons stated in Section 1.1.

B Claim-to-evidence matrix

Claim	Best evidence in snapshot	What remains
Ordered-disjoint frames yield RCC5 models	Lean normal-form and set lemmas; 36,000-network probe	Clean modular rebuild and corrected theorem map
Full models yield surviving cone signatures	<code>coneScheme_complete</code>	Independent pinned build
Surviving fragment signature yields a model	<code>unf_truth</code> , <code>coneScheme_sound</code>	Independent pinned build
Raw $\forall \text{PO}$ -free SAT is decidable	<code>FP0Free</code> , <code>decidableFSat</code>	Independent pinned build, complexity statement, and release engineering
Concrete set SAT matches abstract SAT	<code>satisfiable_iff_set</code>	Credit prior representation theorems precisely and reproduce a pinned build
Full SAT is decidable from a fixed complete finite scheme	Conditional Lean combinator	Actual bounded scheme and proof of completeness
SAT is in Π_1^0	First-order transcription plus Lutz–Wolter r.e. validity argument	Formal encoding/proof calculus instance in Lean
F6 yields complete finite certificates	Prose sketches / abstract adequacy field	RCC5 adequacy/uniformization into an enumerable finite-code grammar; optional bound for direct search

Claim	Best evidence in snapshot	What remains
Full cone soundness	None; boundary formulas expose the submitted gap and defeat named naive repairs	Joint pair/mosaic or different model construction
No solution is established in the reviewed snapshot	2006 primary source plus supplied corpus	Fresh reproducible 2026 literature review

References

- [1] C. Lutz and F. Wolter. Modal logics of topological relations. *Logical Methods in Computer Science*, 2(2:5):1–41, 2006. [https://doi.org/10.2168/LMCS-2\(2:5\)2006](https://doi.org/10.2168/LMCS-2(2:5)2006).
- [2] I. Duenetsch, H. Wang, and S. McCloskey. A relation-algebraic approach to the Region Connection Calculus. *Theoretical Computer Science*, 255(1–2):63–83, 2001. [https://doi.org/10.1016/S0304-3975\(99\)00123-5](https://doi.org/10.1016/S0304-3975(99)00123-5).
- [3] S. Li and M. Ying. Region Connection Calculus: its models and composition table. *Artificial Intelligence*, 145:121–146, 2003. [https://doi.org/10.1016/S0004-3702\(02\)00372-7](https://doi.org/10.1016/S0004-3702(02)00372-7).
- [4] V. Haarslev, C. Lutz, and R. Moeller. A description logic with concrete domains and a role-forming predicate operator. *Journal of Logic and Computation*, 9(3):351–384, 1999. <https://doi.org/10.1093/logcom/9.3.351>.
- [5] M. Nielsen, G. Plotkin, and G. Winskel. Petri nets, event structures and domains, Part I. *Theoretical Computer Science*, 13(1):85–108, 1981. [https://doi.org/10.1016/0304-3975\(81\)90112-2](https://doi.org/10.1016/0304-3975(81)90112-2).
- [6] G. Winskel. Event structures. In *Petri Nets: Applications and Relationships to Other Models of Concurrency*, LNCS 255, pages 325–392, 1987. https://doi.org/10.1007/3-540-17906-2_31.
- [7] W. Szostak and L. Tendera. FO^2 with one transitive relation is decidable. In *STACS 2013*, LIPIcs 20, pages 317–328, 2013. <https://doi.org/10.4230/LIPIcs.STACS.2013.317>.
- [8] L. de Moura and S. Ullrich. The Lean 4 theorem prover and programming language. In *CADE 28*, LNCS 12699, pages 625–635, 2021. https://doi.org/10.1007/978-3-030-79876-5_37.
- [9] M. Wessel et al. ALCI-RCC5 research repository, archived commit 795b2707b72e927c56b465ef0413b77aeb9a66a6, 2026. <https://github.com/lambda-mikel/alcircc5/tree/795b2707b72e927c56b465ef0413b77aeb9a66a6>.